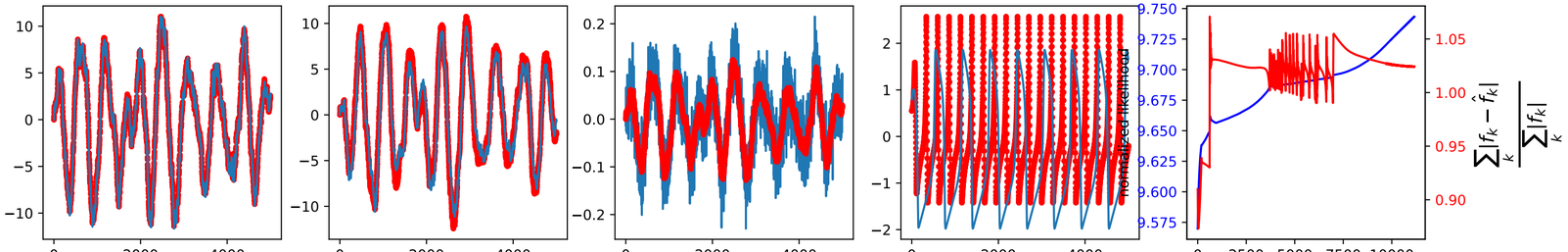
normalized likelihood
$\frac{\sum_k |f_k - \hat{f}_k|}{\sum_k |f_k|}$